

Question	Marking Guidance	Mark	Comments
04.1	Measure temperature (in tube) at intervals and use appropriate corrective measure (if temperature has fluctuated);	1	Accept use thermometer/probe/ equivalent device for measure temperature
04.2	1. Length and diameter OR Surface area OR Volume OR Mass/weight (of cylinders); 2. Time in solution;	2	1. Ignore shape/size 2. Ignore 'time' if unqualified
04.3	1. 1; 2. Use instrument with closer/finer/smaller intervals/graduations/scale;	2	2. Accept correct numerical figure, eg <2 (cm ³) 2. Ignore 'higher resolution'
04.4	1. Water/25°C caused <u>no</u> damage/ <u>no</u> pigment release (in E); 2. (Damage to) <u>cell</u> (-surface) membrane; 3. Ethanol/acid caused some/similar/identical damage OR 70°C caused most damage; 4. (By) ethanol dissolving phospholipid bilayer OR (By acid) altering membrane protein; 5. (By) 70°C denaturing/altering membrane protein OR (By) 70°C increasing fluidity/permeability of membrane;	4 max	1. Accept no colour/intensity change for 'pigment' Accept 'high temperature' for 70°C 1. and 3. Accept description of 'pigment release' for 'damage' 2. Accept description of membrane, eg phospholipid bilayer/bilayer 4. and 5. Accept carrier OR channel OR intrinsic OR extrinsic protein for membrane protein 5. Ignore reference to enzymes unless qualified as membrane bound

TOTAL		9	
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